

Erdős Problem #108

EP-108

L1 - TRACTABLE

OPEN

Graph Theory

ORIGINAL PROBLEM STATEMENT

For every $r \geq 4$ and $k \geq 2$ is there some finite $f(k, r)$ such that every graph of chromatic number $\geq f(k, r)$ contains a subgraph of girth $\geq r$ and chromatic number $\geq k$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>